The Window Effect

Introduction / Idea / Observation

There is an illusion at the Taj Mahal, and at some other places in the world, where if you stand in front of the gate and walk towards the Taj Mahal, it appears to become smaller in size. This is the opposite of what we normally expect. The reverse happens when you walk away from it, it appears to grow.

I have seen this in some famous videos, and I also noticed it myself from the window of my previous apartment, from where a school was visible.

People say this illusion happens because of the architecture, or because the building was made in a special way. I do not think that is the real reason. Even an illusion follows some logic.

My Theory

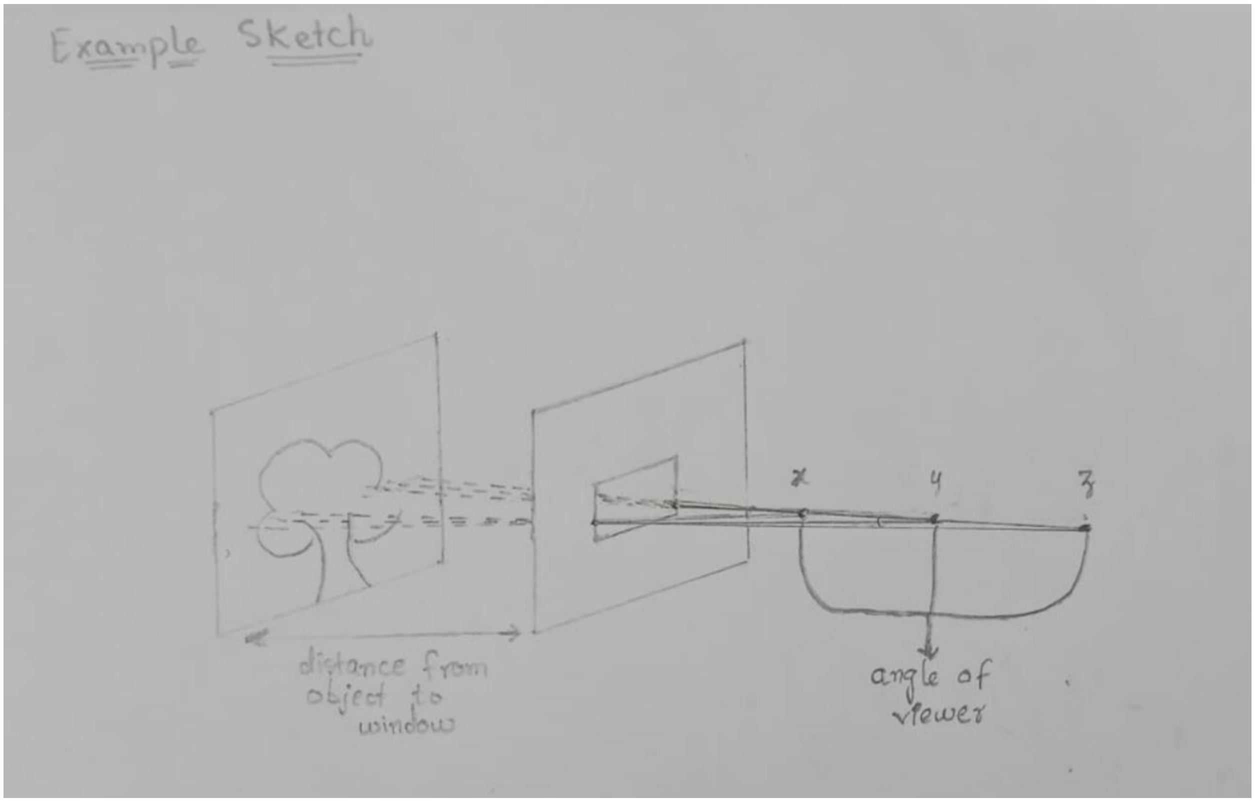
My theory is that this effect is caused by the window or the door. Even a gap between trees can act like a window and create the same thing. This is what I am going to explain, and I call it the **"Window Effect"**.

Experiment

If a window is placed (for example, 3 cm × 3 cm) in front of an object (a scene or a building). From this window, observe the viewer's angle of view at three points:

- Point **x**: 2 cm from the window
- Point **y**: 5 cm from the window
- Point **z**: 7 cm from the window

The distance between the object and the window is kept fixed at 5 cm.



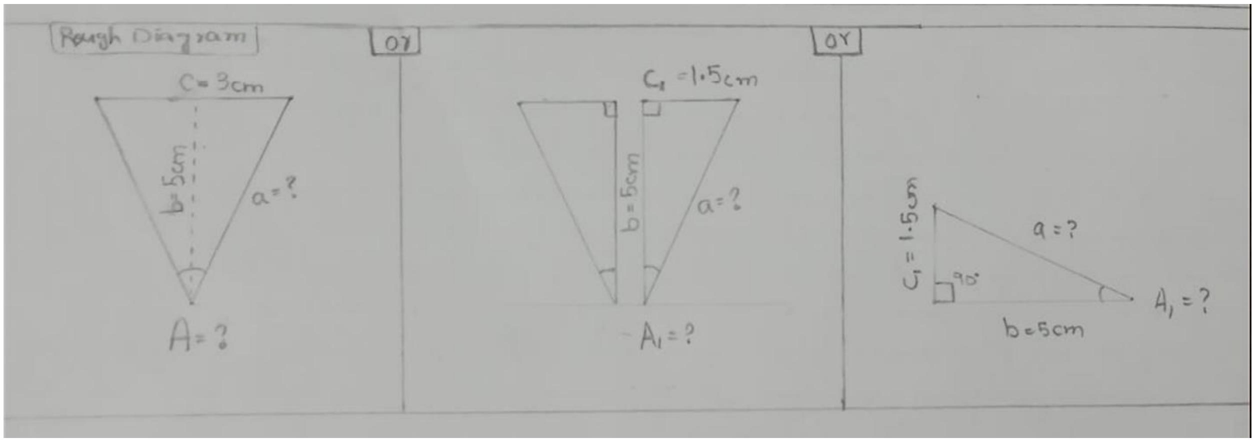
By the end of this practical, I will show why the illusion happens, how to calculate the width of the object (the view) that a viewer can actually see from a point, and how to calculate the angle of view.

Mathematical Approach

Part 1: Finding the angle of view from a distance

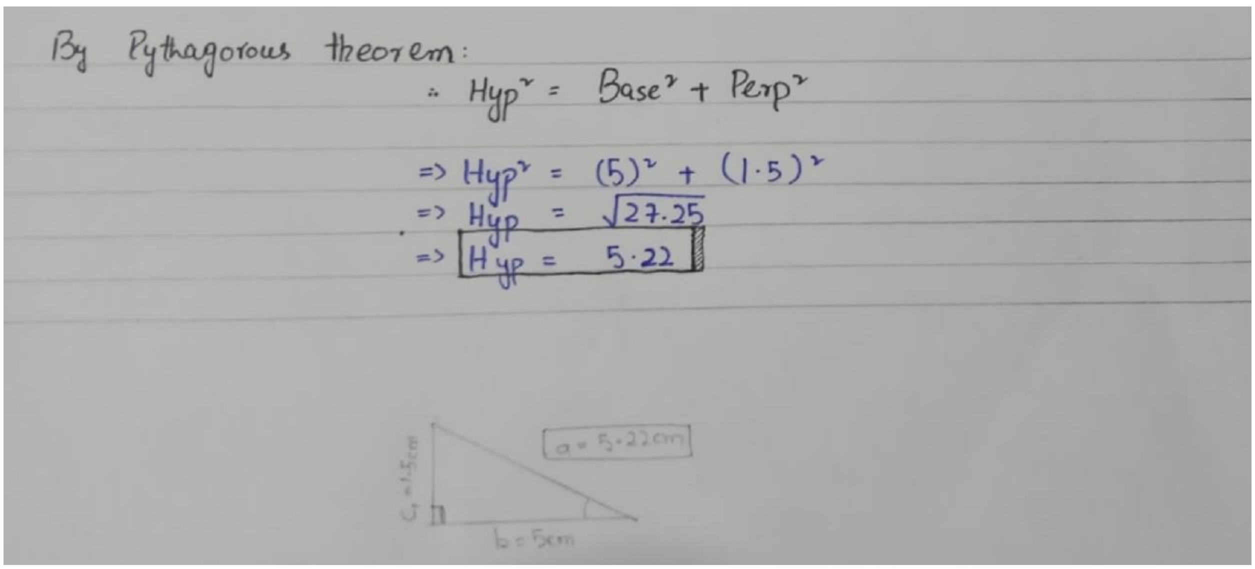
Here I find the angle of view from point **y**, which is 5 cm from the window.

First I divide the triangle into two equal halves, because the window is symmetrical. This gives a right-angled triangle.



To find the angle A_1 using cosine, I first need the hypotenuse (side "a"), because:

$\cos(A_1)$ = Base / Hypotenuse = b / a



Then I use the Pythagoras theorem to find the hypotenuse, then take the inverse cosine to get the half-angle, and finally double it to get the full angle of view.

To calculate angle (A_1):

$$\therefore \cos(A_1) = \frac{\text{Base}}{\text{Hyp}}$$

$$\Rightarrow A_1 = \cos^{-1}\left(\frac{\text{Base}}{\text{Hyp}}\right)$$

$$\Rightarrow A_1 = \cos^{-1}(5/5.22)$$

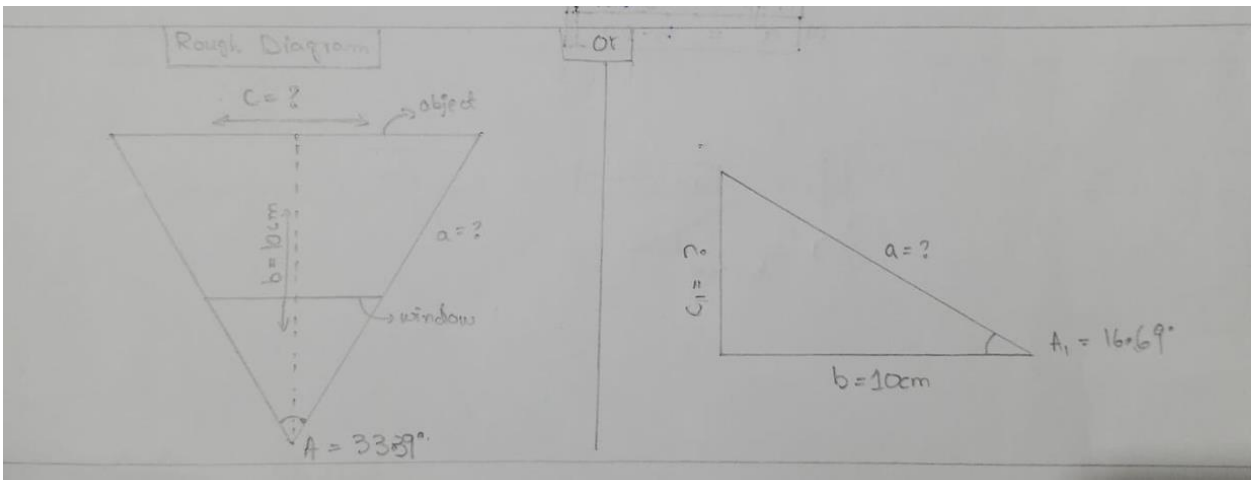
$$\Rightarrow \boxed{A_1 = 16.69^\circ} \longrightarrow \text{for half triangle}$$

$$\Rightarrow A = 2A_1$$

$$\Rightarrow \boxed{A = 33.39^\circ}$$

Part 2: Finding the width of the view seen from a position

Here I calculate the width of the object (y_1) that a viewer can see from position **y**.



To find side "c" (the visible width), I first find side "a" (the hypotenuse) using the angle from Part 1.

$$\therefore \cos(A_1) = \frac{\text{Base}}{\text{Hyp}}$$

$$\Rightarrow \text{Hyp} = \text{Base} / \cos(A_1)$$
$$\Rightarrow \text{Hyp} = 10 / \cos(16.69)$$
$$\Rightarrow \boxed{\text{Hyp} = 10.44\text{ cm}}$$
$$\text{or} \quad \boxed{a = 10.44\text{cm}}$$

Then I use the Pythagoras theorem again to get the visible width, and double it for the full width.

$$\therefore \text{Hyp}^2 = \text{Base}^2 + \text{Perp}^2$$
$$\Rightarrow \text{Perp}^2 = \text{Hyp}^2 - \text{Base}^2$$
$$\Rightarrow \text{Perp}^2 = (10.44)^2 - (10)^2$$
$$\Rightarrow \text{Perp} = \sqrt{8.99}$$
$$\Rightarrow \boxed{\text{Perp} = 3\text{cm}} \longrightarrow \text{half of triangle}$$
$$\Rightarrow \text{Perp} = 2 \times (\text{Perp})$$
$$\Rightarrow \boxed{\text{Perp} = 6\text{ cm}}$$
$$\text{or} \quad \boxed{c = 6\text{cm}}$$

Part 3: Making a general formula

After the worked example, I turned the steps into two general formulas.

Formula to find the angle of view

Requirements:

- width of window = *w*

- distance from window = *d*

Formula no 1:

$$A = \left\{ \cos^{-1} \left[\frac{d}{\sqrt{d^2 + (w/2)^2}} \right] \right\} \times 2$$

Formula to find the width of the object (the view)

Requirements:

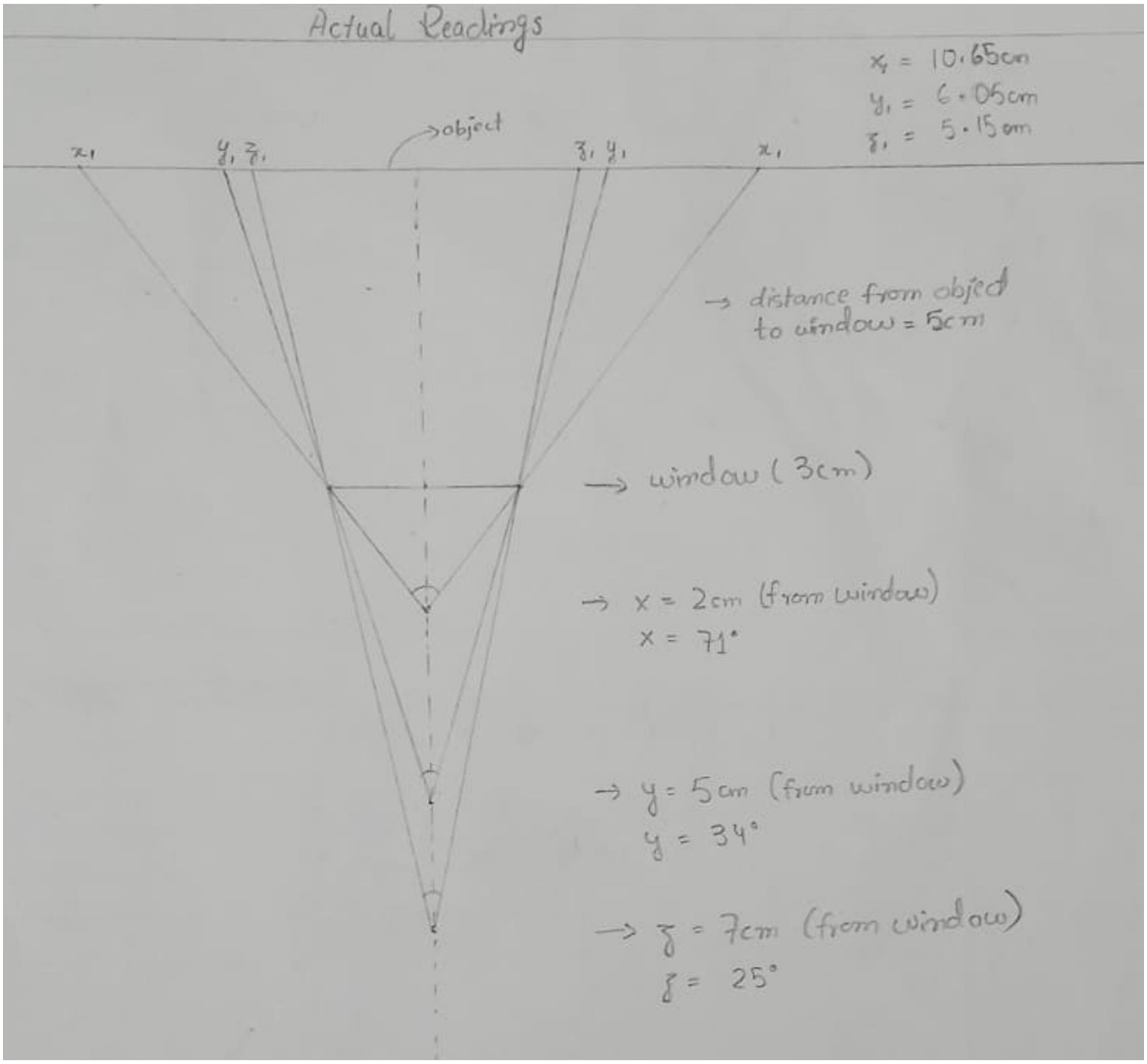
- distance of viewer from object = *d*
- angle of view = *A*

Formula no 2:

$$w = \left\{ \sqrt{\left[\frac{d}{\cos(A/2)} \right]^2 - (d)^2} \right\} \times 2$$

Actual Readings (to compare with the mathematical ones)

I also took real readings from the model, so I could compare them with the calculated values.



Angle of view at each point:

- x = 2 cm → angle = 71°
- y = 5 cm → angle = 34°
- z = 7 cm → angle = 25°

Width of the view seen at each point:

- x_1 = 10.65 cm
- y_1 = 6.05 cm
- z_1 = 5.15 cm

The calculated values and the measured values agree very closely. This shows that the theory is correct.

Why the illusion happens

The window is a fixed frame. Its size never changes.

When you stand close to the window, your angle of view is wide. So you see a wide slice of the object squeezed inside that small frame. This makes the object look zoomed out and small.

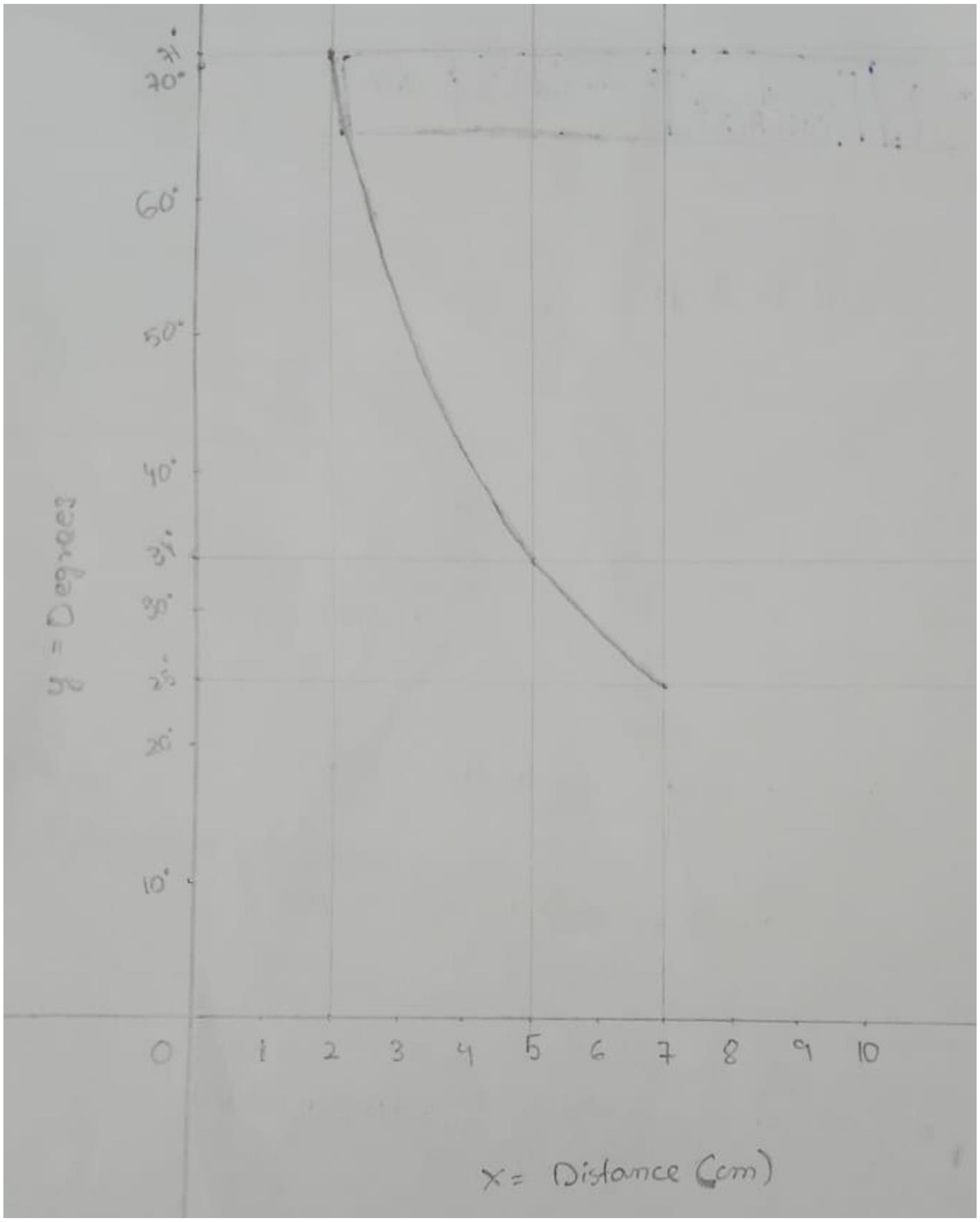
When you stand far from the window, your angle of view is narrow. So you see only a small slice of the object filling the same frame. This makes the object look zoomed in and big.

So as you walk closer, the object appears to shrink, and as you walk back, it appears to grow. This is the Window Effect.

Cases where this effect does not apply

1. When there is no window between the object and the viewer.
2. When there is no distance between the object and the window. In that case the view looks the same from every distance.
3. When the viewer is right at the window, with zero distance from it.

Graph



The graph shows that as the distance from the window increases, the angle of view decreases. The two are inversely related, and the curve falls in a way that looks like a logarithmic decline.